

Python Programming: Lab Activity 1

Due Date: TBD

Objective

This lab focuses on developing basic Python programming skills, including string formatting, print statements, and generating patterns. The exercises are designed to introduce Python's fundamental concepts and strengthen your ability to handle text formatting, loops, and conditional logic.

Assignment Tasks

1. Twinkle Twinkle in a Specific Format

Description: Write a Python program to display the nursery rhyme "Twinkle, Twinkle, Little Star" in a specific format.

Expected Output:

```
Twinkle, twinkle, little star,  
    How I wonder what you are!  
        Up above the world so high,  
        Like a diamond in the sky.  
Twinkle, twinkle, little star,  
    How I wonder what you are
```

2. Display a Statement Across Two Lines

Description: Write a Python program to display the following text across two lines.

Expected Output:

```
I am using Python  
It's my First Assignment
```

3. Statement with Two Methods

Description: Write a Python program to display the following text using two different methods to achieve the same result.

Expected Output:

```
ohhh!!!  
Python is so fun!!! && It is Easy! Get Started
```

4. Generate Patterns

4.1. Pyramid Pattern

Write a Python program to generate the following pyramid pattern of the letter "A":

```
  A
 A A
A A A
A A A A
A A A A A
```

4.2. Box Border Pattern

Write a Python program to generate the following box border pattern using the letter "O":

```
O O O O O
O       O
O       O
O       O
O       O
O O O O O
```

4.3. Stair-Step Pattern

Write a Python program to produce the following stair-step pattern:

```
*
* *
* * *
* * * *
* * * * *
```

4.4. Alphabetic Pyramid

Produce the following pattern using an **alternate method** to what was used for the stair-step pattern:

```
A
B C
D E F
G H I J
K L M N O
```

Submission Guidelines

1. A template code has been provided to you called lab1.py
2. Please do not change anything in the template and add code in designated areas
3. You have been provided instructions to replace a section with your code and uncomment the others.
4. Submit the final file and do not rename it.

Grading Criteria

- **AutoGrader:** Your solution will be evaluated using an autograder, so ensure your output matches the specifications in the question exactly. Partial correctness will not earn marks.
- **Completeness:** Each task is solved as per the instructions.
- **Accuracy:** Outputs match the expected format precisely.
- **Code Quality:** Clear, well-documented, and efficient solutions.
- **Originality:** Use your understanding to complete tasks—plagiarism will result in penalties.